

Akira Ransomware – Practical IV
04-28-2025
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CAP6137 – Malware Reverse Engineering

Executive Summary

This malware was found to be a ransomware developed by the Akira group. It encrypts most files with an extension “.akira”. It excludes all exe and dll files. It leaves behind a readme called “akira_readme.txt” in the directories it does its encryption. It creates threads to carry out its tasks, including the main encryption. This malware is not packed at all, but needs admin rights to execute fully.

The note states that they will analyze the company’s files and come up with a ransom amount. They ask to be contacted at their .onion site, and provide a chatroom key to talk with them.

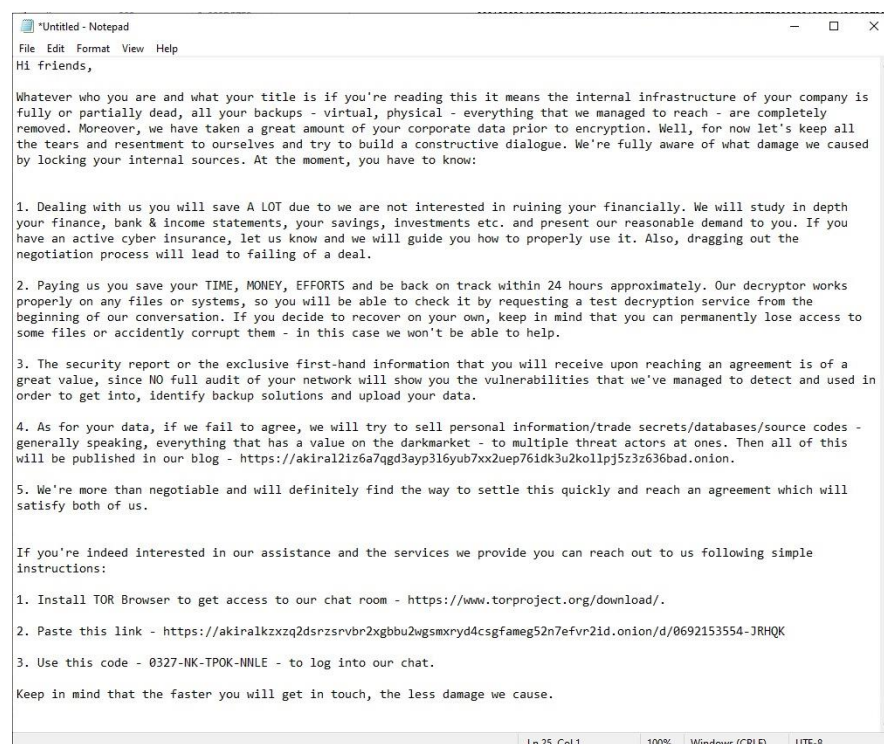
Code analysis

Static analysis:

Sample compiled on: 09/21/2024

compiler-stamp	Wed Aug 21 11:09:01 2024 UTC
debugger-stamp	Wed Aug 21 11:09:01 2024 UTC

Found a ransom-note in strings. Copied its value from PE studio, and pasted on to notepad. Replaced escaped characters with spaces and new lines to make the string more readable.



Here, they let the victim (targeting corporations as they mentioned ‘company’) know that all their data has been stolen and encrypted, and a ransom amount will be communicated to them upon further analysis of the company. They ask the victim to contact them at their onion website (address shown in

the screenshot above). They mention that if the negotiations fail, they will publish all of the company's data.

These strings indicate that a log may be getting created, and on the bottom is also the possible name of the ransom-note (akira_readme.txt).

ascii	39	0x000DEAD8	-	format-string	-	**** LOG ERROR #%04zu ****] [%s] [%s] %s
ascii	21	0x000DA3F0	-	format-string	-	Log-%d-%m-%Y-%H-%M-%S
ascii	17	0x000DEAC0	-	format-string	-	%Y-%m-%d %H:%M:%S
ascii	12	0x000CE5E6	-	format-string	-	xxoJ%r\...\$8
ascii	9	0x000CE1E8	-	format-string	-	J%o%\r8
ascii	9	0x000CE9E7	-	format-string	-	x%oJ%r\.
ascii	8	0x000CEDE8	-	format-string	-	%%oJ.\r\
unicode	49	0x0BF311A0	-	file	-	minkernel\crt\ucrt\inc\corecrt_internal_strtox.h
ascii	16	0x000F6A10	-	file	-	akira_readme.txt

The presence of lengthy imports, and entropy below 7 in all sections indicates that there is no packing or obfuscation in this malware.

property	value	value	value	value	value	value	value
general							
name	.text	.rdata	.data	.pdata	.RDATA	.rsrc	.reloc
md5	6437AEC8CEFA37EA0900BD5...	DF9C8A8450145E38D539A01...	5FB8389E35E5420EBCB0AFA...	80A9BD19510D3A929D96E3F...	5068AA083D3668AED1F99C...	910FFF19CF479C05A7727AD...	EE83E18E64E4178CED7D3A6...
entropy	6.456	5.176	4.882	5.993	3.276	4.718	5.400
file-ratio (99.90%)	77.37 %	15.77 %	3.14 %	3.05 %	0.05 %	0.05 %	0.48 %
raw-address	0x0000400	0x000CB400	0x000F4A00	0x000FCE00	0x00104E00	0x00105000	0x00105200
raw-size (1073664 bytes)	0x000CB000 (831488 bytes)	0x00029600 (169472 bytes)	0x00008400 (33792 bytes)	0x00008000 (32768 bytes)	0x00000200 (512 bytes)	0x00000200 (512 bytes)	0x00001400 (5120 bytes)
virtual-address	0x00001000	0x000CC000	0x000F6000	0x00101000	0x00109000	0x0010A000	0x0010B000
virtual-size (1080308 bytes)	0x000CAFAE (831406 bytes)	0x00029452 (169042 bytes)	0x0000A1AC (41388 bytes)	0x00007F80 (32640 bytes)	0x0000013C (348 bytes)	0x000001E0 (480 bytes)	0x0000138C (5004 bytes)

There are no sections whose presence raises red flags.

There were a lot of file related imports, indicating traversal and manipulation of directories.

WriteFile	✗	0x00000000000F49F4	0x00000000000F49F4	1573 (0x0625)	file
DeleteFileW	✗	0x00000000000F4A88	0x00000000000F4A88	281 (0x0119)	file
SetFileAttributesW	✗	0x00000000000F4AFA	0x00000000000F4AFA	1325 (0x052D)	file
FindFirstFileW	✗	0x00000000000F4F8A	0x00000000000F4F8A	393 (0x0189)	file
FindFirstFileExW	✗	0x00000000000F4F9C	0x00000000000F4F9C	388 (0x0184)	file
FindNextFileW	✗	0x00000000000F4FB0	0x00000000000F4FB0	405 (0x0195)	file
GetFileInformationByHandleEx	✗	0x00000000000F5010	0x00000000000F5010	594 (0x0252)	file
PostQueuedCompletionStatus	-	0x00000000000F48F0	0x00000000000F48F0	1064 (0x0428)	file
ReadFile	-	0x00000000000F49A6	0x00000000000F49A6	1145 (0x0479)	file
GetFileSizeEx	-	0x00000000000F49B2	0x00000000000F49B2	598 (0x0256)	file
GetQueuedCompletionStatus	-	0x00000000000F4A28	0x00000000000F4A28	724 (0x02D4)	file
SetFileInformationByHandle	-	0x00000000000F4A6A	0x00000000000F4A6A	1328 (0x0530)	file
CreateFileW	-	0x00000000000F4AD6	0x00000000000F4AD6	206 (0x00CE)	file
GetFileAttributesW	-	0x00000000000F4AE4	0x00000000000F4AE4	591 (0x024F)	file
GetFileAttributesA	-	0x00000000000F4BC4	0x00000000000F4BC4	586 (0x024A)	file

Also contains a lot of thread related imports

GetCurrentThreadId	x	0x00000000000F4A44	0x00000000000F4A44	549 (0x0225)	execution	i
SleepEx	x	0x00000000000F4AA4	0x00000000000F4AA4	1426 (0x0592)	execution	i
OpenProcess	x	0x00000000000F4C24	0x00000000000F4C24	1042 (0x0412)	execution	i
SetEnvironmentVariableW	x	0x00000000000F53EA	0x00000000000F53EA	1316 (0x0524)	execution	i
GetEnvironmentStringsW	x	0x00000000000F53B6	0x00000000000F53B6	577 (0x0241)	execution	i
TerminateProcess	x	0x00000000000F5132	0x00000000000F5132	1438 (0x059E)	execution	i
TerminateThread	-	0x00000000000F492A	0x00000000000F492A	1439 (0x059F)	execution	i
TlsAlloc	-	0x00000000000F493C	0x00000000000F493C	1456 (0x05B0)	execution	i
TlsFree	-	0x00000000000F498A	0x00000000000F498A	1457 (0x05B1)	execution	i
TlsSetValue	-	0x00000000000F49D6	0x00000000000F49D6	1459 (0x05B3)	execution	i
TlsGetValue	-	0x00000000000F4AAE	0x00000000000F4AAE	1458 (0x05B2)	execution	i
GetCommandLineW	-	0x00000000000F4B58	0x00000000000F4B58	480 (0x01E0)	execution	i
Sleep	-	0x00000000000F4BAA	0x00000000000F4BAA	1423 (0x058F)	execution	i
GetCurrentProcess	-	0x00000000000F4C00	0x00000000000F4C00	544 (0x0220)	execution	i
GetProcessId	-	0x00000000000F4C14	0x00000000000F4C14	704 (0x02C0)	execution	i
FreeEnvironmentStringsW	-	0x00000000000F53D0	0x00000000000F53D0	435 (0x01B3)	execution	i
GetCommandLineA	-	0x00000000000F53A4	0x00000000000F53A4	479 (0x01DF)	execution	i
GetExitCodeThread	-	0x00000000000F4F56	0x00000000000F4F56	583 (0x0247)	execution	i
CreateThread	-	0x00000000000F51CC	0x00000000000F51CC	245 (0x00F5)	execution	i
ExitThread	-	0x00000000000F51DC	0x00000000000F51DC	360 (0x0168)	execution	i
ExitProcess	-	0x00000000000F521C	0x00000000000F521C	359 (0x0167)	execution	i

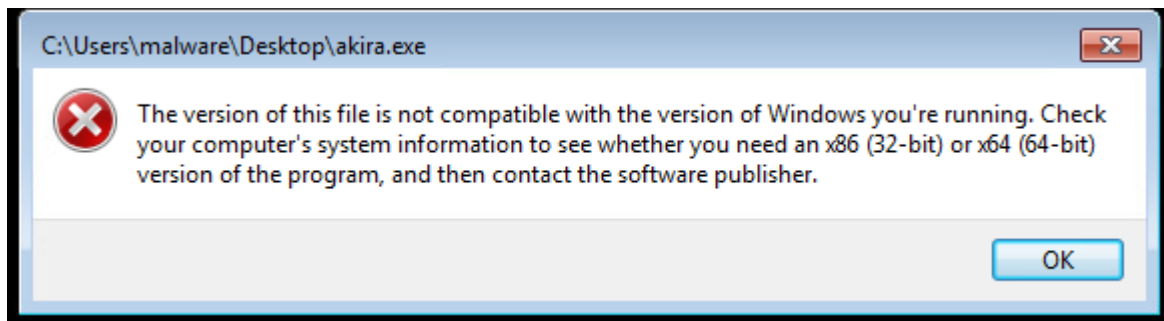
Presence of `IsDebuggerPresent` and `sleep` shows there were possible anti debugging measures taken.

IsDebuggerPresent	-	0x00000000000F5162	0x00000000000F5162	901 (0x0385)	reconnaissance	implicit
Sleep	-	0x00000000000F4BAA	0x00000000000F4BAA	1423 (0x058F)	execution	implicit

Dynamic Analysis:

The sample was renamed to `akira.exe` for ease of analysis.

It must be noted that the malware can not run on windows 7.



Therefore, analysis was done on windows 10 machine. The malware was initially run in user mode, but that resulted in a lot of “access denied” errors when creating files.

3:57:2...	akira.exe	7720	CreateFile	C:\akira_readme.txt	ACCESS DENIED	Desired Access: G...
3:57:5...	akira.exe	7720	CreateFile	C:\\$WinREAgent\akira_readme.txt	ACCESS DENIED	Desired Access: G...
4:31:4...	akira.exe	7720	CreateFile	C:\\$WinREAgent\Scratch\akira_readm...	ACCESS DENIED	Desired Access: G...
4:32:1...	akira.exe	7720	CreateFile	C:\Config.Msi\akira_readme.txt	ACCESS DENIED	Desired Access: G...
4:32:4...	akira.exe	7720	CreateFile	C:\Config.Msi	ACCESS DENIED	Desired Access: R...
4:33:0...	akira.exe	7720	CreateFile	C:\Users\akira_readme.txt	ACCESS DENIED	Desired Access: G...
4:33:1...	akira.exe	7720	CreateFile	C:\Documents and Settings	ACCESS DENIED	Desired Access: R...

Therefore, the code was run again in admin mode (in `x64dbg`).

Breakpoints were set on all instances of **ReadFile**, **CreateFile**, **WriteFile**, **DeleteFile**, **IsDebuggerPresent**, **Sleep**, **ReadConsole**, **WriteConsole**, **FindFirstFile** and **FindNextFile**. They were also set on all instances of **WSAStartup** and **WNetGetConnection** to make note of any network activities.

00007FF6A83D11C3	akira.exe	Enabled	call	qword ptr ds:[<&WSAStartup>]	1
00007FF6A83D1483	akira.exe	Enabled	call	qword ptr ds:[<&WSAStartup>]	0
00007FF6A83D1683	akira.exe	Enabled	call	qword ptr ds:[<&WSAStartup>]	0
00007FF6A83D1873	akira.exe	Enabled	call	qword ptr ds:[<&WSAStartup>]	0
00007FF6A83D3673	akira.exe	Enabled	call	qword ptr ds:[<&WSAStartup>]	0
00007FF6A8408509	akira.exe	Enabled	call	qword ptr ds:[<&SleepEx>]	0
00007FF6A8410968	akira.exe	Enabled	call	qword ptr ds:[<&WriteFile>]	0
00007FF6A8410B89	akira.exe	Enabled	call	qword ptr ds:[<&ReadFile>]	0
00007FF6A8412A87	akira.exe	Enabled	call	qword ptr ds:[<&WNetGetConnectionW>]	0
00007FF6A8412BA7	akira.exe	Enabled	call	qword ptr ds:[<&WNetGetConnectionW>]	0
00007FF6A841C595	akira.exe	Enabled	call	qword ptr ds:[<&Sleep>]	0
00007FF6A842C4C4	akira.exe	Enabled	call	qword ptr ds:[<&WriteConsoleA>]	0
00007FF6A842C52A	akira.exe	Enabled	call	qword ptr ds:[<&WriteConsoleA>]	0
00007FF6A842C562	akira.exe	Enabled	call	qword ptr ds:[<&WriteConsoleA>]	0
00007FF6A842C587	akira.exe	Enabled	call	qword ptr ds:[<&WriteFile>]	0
00007FF6A844A3C7	akira.exe	Enabled	call	qword ptr ds:[<&Sleep>]	0
00007FF6A8451BD8	akira.exe	Enabled	call	qword ptr ds:[<&FindNextFileW>]	3
00007FF6A8451C57	akira.exe	Enabled	call	qword ptr ds:[<&FindFirstFileExW>]	1
00007FF6A8451D56	akira.exe	Enabled	call	qword ptr ds:[<&FindFirstFileW>]	0
00007FF6A8451FEB	akira.exe	Enabled	call	qword ptr ds:[<&CreateFileW>]	0
00007FF6A845D724	akira.exe	Enabled	call	qword ptr ds:[<&IsDebuggerPresent>]	0
00007FF6A8463091	akira.exe	Enabled	call	qword ptr ds:[<&LoadLibraryExW>]	2
00007FF6A84630C9	akira.exe	Enabled	call	qword ptr ds:[<&LoadLibraryExW>]	0
00007FF6A846432D	akira.exe	Enabled	call	qword ptr ds:[<&IsDebuggerPresent>]	0
00007FF6A84736E3	akira.exe	Enabled	call	qword ptr ds:[<&LoadLibraryExW>]	8
00007FF6A8473733	akira.exe	Enabled	call	qword ptr ds:[<&LoadLibraryExW>]	0
00007FF6A847685A	akira.exe	Enabled	call	qword ptr ds:[<&WriteFile>]	0
00007FF6A84768AC	akira.exe	Enabled	call	qword ptr ds:[<&WriteFile>]	0
00007FF6A8476AD3	akira.exe	Enabled	call	qword ptr ds:[<&WriteFile>]	2
00007FF6A8476BEE	akira.exe	Enabled	call	qword ptr ds:[<&WriteFile>]	0
00007FF6A8476D57	akira.exe	Enabled	call	qword ptr ds:[<&WriteFile>]	0
00007FF6A84770CE	akira.exe	Enabled	call	qword ptr ds:[<&WriteFile>]	3
00007FF6A8477AAE	akira.exe	Enabled	call	qword ptr ds:[<&ReadFile>]	0
00007FF6A8477D4F	akira.exe	Enabled	call	qword ptr ds:[<&ReadFile>]	0
00007FF6A8477F85	akira.exe	Enabled	call	qword ptr ds:[<&ReadFile>]	0
00007FF6A8478585	akira.exe	Enabled	call	qword ptr ds:[<&ReadConsoleW>]	0
00007FF6A84785D7	akira.exe	Enabled	call	qword ptr ds:[<&ReadFile>]	0
00007FF6A8478816	akira.exe	Enabled	call	qword ptr ds:[<&FindFirstFileExW>]	0
00007FF6A8478928	akira.exe	Enabled	call	qword ptr ds:[<&FindNextFileW>]	0
00007FF6A8482672	akira.exe	Enabled	call	qword ptr ds:[<&CreateFileW>]	4
00007FF6A84826C2	akira.exe	Enabled	call	qword ptr ds:[<&CreateFileW>]	0
00007FF6A84828D4	akira.exe	Enabled	call	qword ptr ds:[<&CreateFileW>]	0
00007FF6A8484172	akira.exe	Enabled	call	qword ptr ds:[<&CreateFileW>]	0
00007FF6A84841DD	akira.exe	Enabled	call	qword ptr ds:[<&WriteConsoleW>]	0
00007FF6A848422C	akira.exe	Enabled	call	qword ptr ds:[<&CreateFileW>]	0
00007FF6A8484248	akira.exe	Enabled	call	qword ptr ds:[<&WriteConsoleW>]	0
00007FF6A848581F	akira.exe	Enabled	call	qword ptr ds:[<&CreateFileW>]	0
00007FF6A8486AE3	akira.exe	Enabled	call	qword ptr ds:[<&CreateFileW>]	0
00007FF6A8486B38	akira.exe	Enabled	call	qword ptr ds:[<&CreateFileW>]	0
00007FF6A8489A84	akira.exe	Enabled	call	qword ptr ds:[<&DeleteFileW>]	0

The function responsible for iterating through the directory was found to be at offset **0x6e970**. **FindFirstFile** and **FindNextFile** were both backtracked to the function starting at this address in Ghidra.

00007FF6A841C2A	48:802A	mov rax,rax	rsi:L"C:\\\$WinREAgent\\", rcx:L"C:\\\$WinREAgent\\"	RAX	000000ADBDFFAFA0	4C:"C:\\\$WinREAgent\\\"
00007FF6A841C2D	48:8BF1	mov rsi,rcx	rcx:L"C:\\\$WinREAgent\\", rcx:L"C:\\\$WinREAgent\\"	RIP	000000ADBDFFB010	L"C:\\\$WinREAgent\\\"
00007FF6A841C30	74 00	je akira.7FF6A8451C3F		RCX	0000002525B0A110	
00007FF6A841C32	48:8B0A	mov rcx,qword ptr ds:[rdx]		RBP	000000ADBDFFB0C0	
00007FF6A841C35	FF15 C5A60400	call qword ptr ds:[<@indtose>]		RSP	000000ADBDFFB0D0	
00007FF6A841C38	8BC0	test eax,esi		RST	0000002525B0A110	
00007FF6A841C3D	74 41	je akira.7FF6A8451C80		RDI	000000ADBDFFB180	
00007FF6A841C3F	8B424 28 00	and qword ptr rsi:[rsp+28],0				
00007FF6A841C44	45:33C9	xor rdi,rdi				
00007FF6A841C47	48:8B424 20 00	and qword ptr rsi:[rsp+20],0				
00007FF6A841C4D	4C:8BC7	mov rdi,rdi				
00007FF6A841C50	41:80C2	mov rcx,rsi				
00007FF6A841C53	41:8051 01	test eax,qword ptr ds:[rcx]				
00007FF6A841C5D	FF15 83A60400	call qword ptr ds:[<@indtose>]				
00007FF6A841C60	48:8903	mov rcx,qword ptr ds:[rcx]				
00007FF6A841C63	48:898F FF	cmp rcx,FFFFFFFFFFFFFFFF				
00007FF6A841C66	74 04	je akira.7FF6A8451C8A				
00007FF6A841C68	3BC0	cmp rcx,FFFFFFFFFFFFFFFF				
00007FF6A841C6B	1B 06	je akira.7FF6A8451C70				
00007FF6A841C6A	FF15 83A60400	call qword ptr ds:[<@indtose>]				
00007FF6A841C70	48:8B5C24 40	mov rcx,qword ptr rsi:[rsp+40]				

0016e970 48 89 5c ...	MOV	qword ptr [RSP + local_res10], RBX	47	tnumk_HUN_UULW(ZUPpvvar);
0016e975 56	PUSH	RDI	48	}
0016e976 57	PUSH	RDI	49	if (7 < *(ulonglong *) (param_1 + 0xc)) {
0016e977 41 56	PUSH	RDI	50	param_1 = *(LPWSTR *) param_1;
0016e979 48 83 ec 50	SUB	RSP, 0x50	51	}
0016e97d 49 8b d0	MOV	RBX, param_4	52	DVar4 = _std_fs_directory_iterator_open(param_1,param_3,param_4);
0016e980 4d 8b f0	MOV	R14, param_3	53	if (DVar4 == 0) {
0016e983 8b f2	MOV	ESI, param_2	54	pVar1 = (HANDLE) * param_3;
0016e985 48 8b f9	MOV	RDI, param_1	55	uVar6 = FUN_0016d800((longlong) param_4);
0016e988 48 83 79 ...	CMPC	qword ptr [param_1 + 0x18], 0x8	56	cVar3 = (char) uVar6;
0016e98d 72 03	JNC	LAB_0016e992	57	while(true) {
0016e98f 48 8b 09	MOV	param_1, qword ptr [param_1]	58	if (cVar3 == '\0') {
			59	return 0;
			60	}
			61	DVar4 = funcfindnextfiles(pvVar1,param_4);
			62	if (DVar4 != 0) break;
			63	uVar6 = FUN_0016d800((longlong) param_4);
			64	cVar3 = (char) uVar6;
			65	}
			66	}

_std_fs_directory_iterator_open eventually calls **FindFirstFile** and **funcfindnextfiles** (renamed for ease of analysis) are both found at function at address **0x6e970**.

We also see the function responsible for creating the readme at offset **0xb256c** (Renamed to ReadMeMaker for analysis) (Ghidra base set to 0x7ff6a83d0000).

```

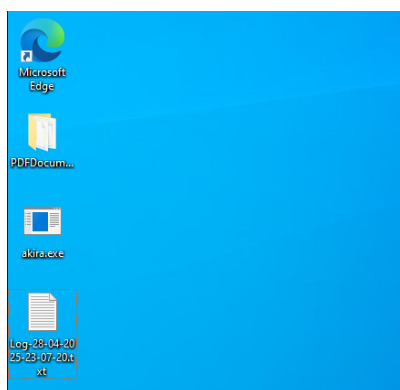
7ff6a84256c 48 8b c4      MOV     RAX, RSP
7ff6a84256f 48 8b 5b      MOV     RDI, RSI
7ff6a842572 48 8b 70      MOV     RDI, RSI
7ff6a842577 48 8b 78      MOV     RDI, RSI
7ff6a84257b 4c 89 40 18   MOV     RAX, RDI
7ff6a84257f 35          PUSH    RBP
7ff6a842580 41 54        PUSH    R13
7ff6a842582 41 55        PUSH    R13
7ff6a842584 41 56        PUSH    R14
  
```

and it is written at offset **0x42410** which was backtracked and found in x32dbg and confirmed in Ghidra

```

7ff6a842410 48 8b c4      MOV     RAX, RSP
7ff6a842411 48 8b 5b      MOV     RDI, RSI
7ff6a842416 48 8b 70      MOV     RDI, RSI
7ff6a842417 48 8b 78      MOV     RDI, RSI
7ff6a84241b 4c 89 40 18   MOV     RAX, RDI
7ff6a84241f 35          PUSH    RBP
7ff6a842420 41 54        PUSH    R13
7ff6a842422 41 55        PUSH    R13
7ff6a842424 41 56        PUSH    R14
  
```

The log file previously mentioned is created on Desktop



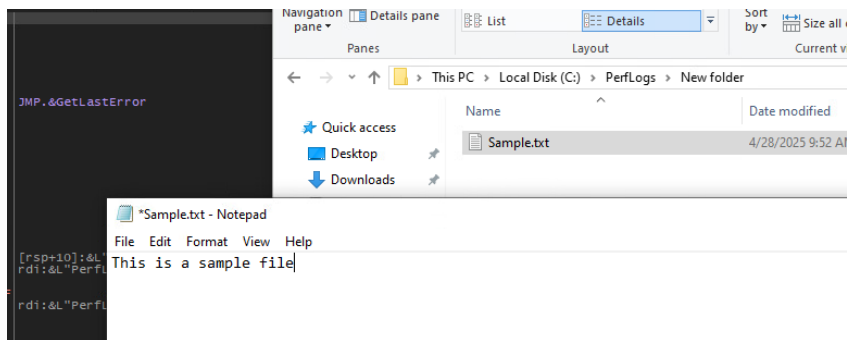
The beginning few lines indicate a very important behavior

```

Log-28-04-2025-23-07-20.txt - Notepad
File Edit Format View Help
[2025-04-28 23:07:21.026] [file_logger] [info] Number of thread to folder parsers = 1
[2025-04-28 23:07:21.026] [file_logger] [info] Number of thread to root folder parsers = 1
[2025-04-28 23:07:21.026] [file_logger] [info] Number of threads to encrypt = 2
[2025-04-28 23:07:21.027] [file_logger] [error] File handle not found! (C:\Program Files\7 Zip\7
  
```

This indicates that the malware is making use of threads to carry out its tasks. This was verified by manually monitoring the files being encrypted.

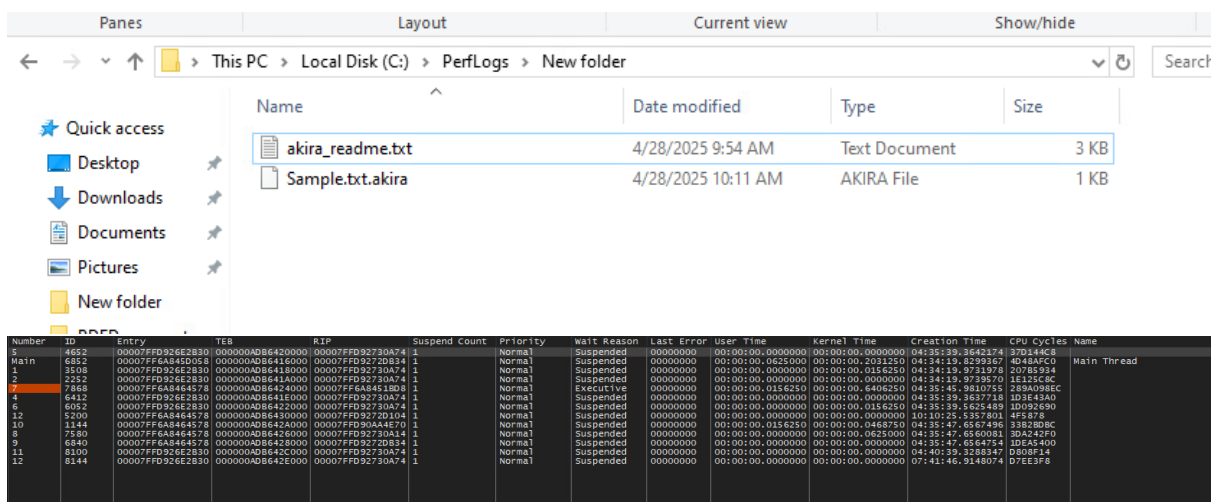
A sample file was added to study the encryption pipeline of the malware.



The malware then detected the file.



There was NO exclusive write on the file through the main malware. In fact, it started reading further files before it encrypted the sample. It was only after a while that the sample was encrypted with a .akira extension, and there was some activity noticed in the threads section of x64dbg.



Running the malware also makes use of ncrpt.dll from system32 which was probably used for encrypting the files.

12:03:...	akira.exe	3472	CreateFile	C:\Windows\System32\ncrypt.dll	SUCCESS	Desired Access: R...
12:03:...	akira.exe	3472	QueryBasicInfo...	C:\Windows\System32\ncrypt.dll	SUCCESS	CreationTime: 3/26...
12:03:...	akira.exe	3472	CloseFile	C:\Windows\System32\ncrypt.dll	SUCCESS	
12:03:...	akira.exe	3472	CloseFile	C:\Windows\System32\ncrypt.dll	SUCCESS	
12:03:...	akira.exe	3472	CreateFile	C:\Windows\System32\ncrypt.dll	SUCCESS	Desired Access: R...
12:03:...	akira.exe	3472	CreateFileMapp...	C:\Windows\System32\ncrypt.dll	FILE LOCKED WI...	SyncType: SyncTy...
12:03:...	akira.exe	3472	CreateFileMapp...	C:\Windows\System32\ncrypt.dll	SUCCESS	SyncType: SyncTy...
12:03:...	akira.exe	3472	Load Image	C:\Windows\System32\ncrypt.dll	SUCCESS	Image Base: 0x7f8...
12:03:...	akira.exe	3472	QueryNameInfo...	C:\Windows\System32\ncrypt.dll	SUCCESS	Name: \Windows\...
12:03:...	akira.exe	3472	CreateFile	C:\Windows\System32\ncrypt.dll	SUCCESS	Desired Access: G...
12:03:...	akira.exe	3472	CloseFile	C:\Windows\System32\ncrypt.dll	SUCCESS	

No major changes were noticed in the registry

[illegible]

There were some keys added to VSS, but nothing indicates its deletion.

One important observation about the malware is that it encrypted all the files for C:\Users\Default, but not all for C:\Users\Malware, and it also did not encrypt any .exe or .dll files.

Before running:

 sample.dll	4/29/2025 2:33 AM	Application exten...	1 KB
 sample.exe	4/29/2025 2:33 AM	Application	1 KB
 sample.txt	4/29/2025 2:33 AM	Text Document	1 KB

After running:

Name	Date modified	Type	Size
 akira_readme.txt	4/29/2025 2:35 AM	Text Document	3 KB
 sample.dll	4/29/2025 2:33 AM	Application exten...	1 KB
 sample.exe	4/29/2025 2:33 AM	Application	1 KB
 sample.txt.akira	4/29/2025 2:35 AM	AKIRA File	1 KB

Network analysis

There was absolutely no network activity observed, even though the ransom note stated that they have all the company data. There were no suspicious DNS requests made, and no direct communication to any IP without Domain Name Resolution other than that of remnux, and the debugger did not break at the network related function calls either. Inetsim, accept-all-ips, fakedns, and wireshark were all used to verify this. (ps in the screenshot, wj32.org might look off but it is the domain of the creator of Process Hacker, Wen Jia Liu ‘wj32’)

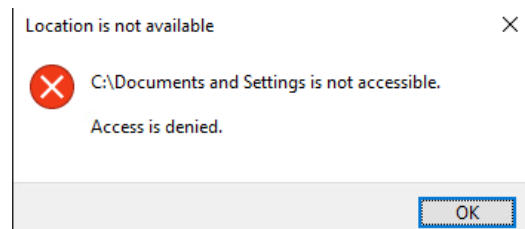
Analysis of Files

Like previously mentioned, it created a log file in the same directory as where it is run.

```
Log-29-04-2025-02-35-01.txt - Notepad
File Edit Format View Help
[2025-04-29 02:35:02.097] [file_logger] [info] Number of thread to folder parsers = 1
[2025-04-29 02:35:02.097] [file_logger] [info] Number of threads to root folder parsers = 1
[2025-04-29 02:35:02.097] [file_logger] [info] Number of threads to encrypt = 2
[2025-04-29 02:35:02.127] [file_logger] [error] Get file size failed! (C:\DumpStack.log.tmp)
[2025-04-29 02:35:02.904] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\Alphabet.xml)
[2025-04-29 02:35:02.905] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\ar-SA\tipresx.dll.mui)
[2025-04-29 02:35:02.908] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\bg-BG\tipresx.dll.mui)
[2025-04-29 02:35:02.908] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\Content.xml)
[2025-04-29 02:35:02.909] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\cs-CZ\tipresx.dll.mui)
[2025-04-29 02:35:02.909] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\da-DK\tipresx.dll.mui)
[2025-04-29 02:35:02.911] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\de-DE\tipresx.dll.mui)
[2025-04-29 02:35:02.914] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\el-GR\tipresx.dll.mui)
[2025-04-29 02:35:02.916] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-GB\tipresx.dll.mui)
[2025-04-29 02:35:02.919] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\inkObj.dll.mui)
[2025-04-29 02:35:02.919] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\inputPersonalization.exe.mui)
[2025-04-29 02:35:02.920] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\micaut.dll.mui)
[2025-04-29 02:35:02.920] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\mlp.exe.mui)
[2025-04-29 02:35:02.922] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\mshwLatin.dll.mui)
[2025-04-29 02:35:02.922] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\ptscm.dll.mui)
[2025-04-29 02:35:02.922] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\ShapeCollector.exe.mui)
[2025-04-29 02:35:02.922] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\tabskb.dll.mui)
[2025-04-29 02:35:02.922] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\TabTip.exe.mui)
[2025-04-29 02:35:02.923] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\tipresx.dll.mui)
[2025-04-29 02:35:02.924] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\TipRes.dll.mui)
[2025-04-29 02:35:02.924] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\en-US\TipTsf.dll.mui)
[2025-04-29 02:35:02.926] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\es-ES\tipresx.dll.mui)
[2025-04-29 02:35:02.927] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\es-MX\tipresx.dll.mui)
[2025-04-29 02:35:02.928] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\es-EE\tipresx.dll.mui)
[2025-04-29 02:35:02.929] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fi-FI\tipresx.dll.mui)
[2025-04-29 02:35:02.930] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fr-CA\tipresx.dll.mui)
[2025-04-29 02:35:02.933] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fr-FR\tipresx.dll.mui)
[2025-04-29 02:35:02.936] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fsdefinitions\auxpad.xml)
[2025-04-29 02:35:02.937] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fsdefinitions\auxpad\auxbase.xml)
[2025-04-29 02:35:02.939] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fsdefinitions\insert\insertbase.xml)
[2025-04-29 02:35:02.939] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fsdefinitions\insert.xml)
[2025-04-29 02:35:02.942] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fsdefinitions\keypad\keybase.xml)
[2025-04-29 02:35:02.942] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fsdefinitions\keypad\keypadbase.xml)
[2025-04-29 02:35:02.943] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fsdefinitions\keypad\kor-kor.xml)
[2025-04-29 02:35:02.943] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fsdefinitions\keypad.xml)
[2025-04-29 02:35:02.946] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fsdefinitions\main\baseAltGr_rtl.xml)
[2025-04-29 02:35:02.947] [file_logger] [error] File handle not found! (C:\Program Files\Common Files\microsoft shared\ink\fsdefinitions\main\base.xml)
```

It created a directory ‘Documents and Settings’ which cannot be accessed by the user.

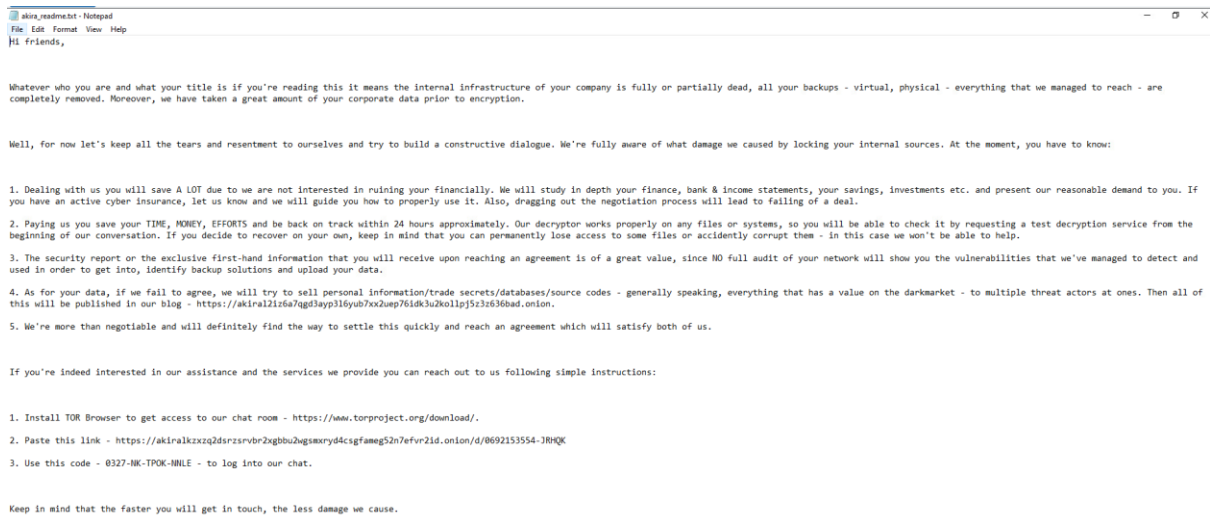
ie ...	Process Name	PID	Operation	Path	Result	Detail
59:...	akira.exe	1476	CreateFile	C:\Documents and Settings	REPARSE	Desired Access: R...
59:...	akira.exe	1476	CreateFile	C:\Users	SUCCESS	Desired Access: R...
59:...	akira.exe	1476	QueryBasicInfor...	C:\Users	SUCCESS	CreationTime: 12/7...
59:...	akira.exe	1476	CloseFile	C:\Users	SUCCESS	



Interestingly, the malware creates certain .arika files as well (notice the difference in name. It is different from .akira). The file names are seemingly random alphanumeric values followed by .arika. However, none of these files can be accessed. These seem to be formed in the process of encryption.

12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.encoding.extensions\4.3.0\ref\xamarinivos10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.encoding.extensions\4.3.0\ref\xamarinwatchos10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\MonoAndroid10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\MonoTouch10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\net45\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\portable-net45+win8+wp8+wp81\77efa9ddf8405a8bceb6a37939b6e...
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\win8\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\wp80\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\wp81\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\xamarinios10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\xamarinmac20\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\xamarinivos10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\lib\xamarinwatchos10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\MonoAndroid10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\MonoTouch10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\net45\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\portable-net45+win8+wp8+wp81\77efa9ddf8405a8bceb6a37939b6e...
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\win8\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\wp80\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\wp81\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\xamarinios10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\xamarinmac20\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\xamarinivos10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.1.0\ref\xamarinwatchos10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\MonoAndroid10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\MonoTouch10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\net45\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\portable-net45+win8+wp8+wp81\77efa9ddf8405a8bceb6a37939b6e...
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\win8\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\wp80\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\wp81\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\xamarinios10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\xamarinmac20\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\xamarinivos10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\lib\xamarinwatchos10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\ref\MonoAndroid10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\ref\MonoTouch10\77efa9ddf8405a8bceb6a37939b6ef1.anka
12:09.....	akira.exe	2640	CreateFile	C:\Program Files (x86)\Microsoft SDKs\NuGetPackages\system.text.regularexpressions\4.3.0\ref\net45\77efa9ddf8405a8bceb6a37939b6ef1.anka

In addition to the .akira files, it also created the readme files.



Indicators of Compromise

1. At the location where the sample is located, one can find a log file created with the date and .
2. Many files will be encrypted with a .akira extension.
3. Presence of akira_readme.txt (ransomnote) in folders where files were encrypted.
4. Files on desktop, and a lot of other files in the running users folder (C:\Users\Malware here) are left unencrypted, but not the ones in C:\Users\Default.

YARA Rule

Apart from strings in the ransomnote and its name, variable threadmaker is the byte sequence around the part where it is creating threads. Also, the rule did not detect any other binary in PMA Lab exercises. (In remnux, the file was renamed to akira for analysis) (Rule also included in a separate txt file with the submission).

```
remnux@remnux:~/Desktop$ cat detectakira.yara
rule detectakira
{
    meta:
        description = "Detects ransomware of the akira family"
        author = "Vandan Agrawal"
        date = "2025-04-28"

    strings:
        $tor = "torproject.org" nocase
        $akira = "akira" nocase
        $extension = ".onion" nocase
        $exec = { 4D 5A }
        $file = "akira_readme.txt" nocase
        $threadmaker = { 4C 8D 05 7D FE FF FF 8B 44 24 60 4C 8B CB 48 8B CE 89 44 24 20 FF 15 D9
7C 03 00 }

    condition:
        $exec at 0 and $tor and $akira and $extension and $file and $threadmaker
}remnux@remnux:~/Desktop$ yara -r detectakira.yara .
detectakira ./GradPractical4Samples/VandanAgarwal/akira
```